

# Non-Homogeneous Spatial Filter Optimization for EEG-based Brain-Computer Interfaces

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**Abstract**—Neuronal power attenuation or enhancement in specific frequency bands over the sensorimotor cortex, called Event-Related Desynchronization (ERD) or Event-Related Synchronization (ERS), respectively, is a major phenomenon in brain activities involved in imaginary movement of body parts. However, it is known that the nature of motor imagery-related electroencephalogram (EEG) signals is non-stationary and highly variable over time and frequency. In this paper, we propose a novel method of finding a discriminative time- and frequency-dependent spatial filter, which we call 'non-homogeneous filter.' We adaptively select bases of spatial filters over time and frequency. By taking both temporal and spectral features of EEGs in finding a spatial filter into account it is beneficial to be able to consider non-stationarity of EEG signals. In order to consider changes of ERD/ERS patterns over the time-frequency domain, we devise a spectrally and temporally weighted classification method via statistical analysis. Our experimental results on the BCI Competition IV dataset II-a clearly presented the effectiveness of the proposed method outperforming other competing methods in the literature.

**Index Terms**—Brain-Computer Interface (BCI); Motor Imagery; Spatial Filter Optimization; Electroencephalogram (EEG)

## I. INTRODUCTION

A Brain-Computer Interface (BCI), which translates human intentions into commands to control an external device by analyzing and recognizing brain signals, allows users to communicate with their environment without using the brain's normal output pathways of peripheral nerves and muscles [2,3]. Therefore, it has been of great interest to many research groups for patients that suffer from amyotrophic lateral sclerosis, locked-in syndrome, etc.

Among various methods of eliciting the specific regulation of brain signals, imagined body-part movement, called motor imagery, has been most widely used in an EEG-based BCI due to its naturalness and spontaneousness. In terms of neurophysiology, motor imagery accompanies attenuation (ERD: Event-Related Desynchronization) or enhancement (ERS: Event-Related Synchronization) of rhythmical synchrony over the sensorimotor cortex in the specific frequency bands of  $\mu$ -rhythm (8~13Hz) and  $\beta$ -rhythm (14~30Hz) [4,5]. However, it is also known that the motor imagery responsive frequency bands are highly variable for inter- and intra-subjects. Furthermore, due to the volume conduction effect in EEG [6,7] finding a class-discriminative spatial filter that certainly improves classification performance

has been considered as a challenging problem in the BCI community [8,9].

While many previous methods showed satisfactory results in their experiments, they did not consider the non-stationary feature of human brain activity [10,11,12]. That is, they found an optimal spatial filter from samples in the predefined time segment and applied it to all the sample points within the time segment homogeneously in classification. We call this type of spatial filter 'homogeneous' spatial filter. On the while, in this paper, we propose a novel method of finding a bank of spatial filters, called 'non-homogeneous' spatial filters, each of which is learned for a different segment in time.

We propose a novel method of finding non-homogeneous spatial filters. Unlike other Common Spatial Patterns (CSP) [14]-based methods that find optimal spatial filters and apply them homogeneously over all samples, the proposed method optimizes spatial filters for each of the time-frequency segments to reflect the non-stationary feature of human brain activity.. From our experiment on a public dataset, it was proved that the proposed weighting scheme improved the classification accuracy.

The rest of the paper is organized as follows. In Section II, we describe a method for non-homogeneous spatial filters optimization and for the construction of a time-frequency weighted classifier. Experimental results are presented in Section III and concluding remarks are drawn in Section IV.

## II. PROPOSED METHOD

### A. Space-Time-Frequency Representation

We decompose EEG signals to 9 frequency bands between 4 Hz and 40 Hz with a 4 Hz window size. For each decomposed signal, we apply CSP spatial filters [13] to get CSP filtered signals. Each source signal is separated to 9 time segment using 50% overlapped 1s size windows. For comparison, we select same band ranges which are used in the Ang et al.'s work [14]. For each separated signal on frequency and time domains, we calculate the variance of source signals to construct space-time-frequency feature space.

### B. Feature Selection on the Feature Space

We apply F-test to each space-time-frequency feature space to select the discriminative features of EEG. F-test is the statistical method to find differences of data sets using the between-class variance and within-class variance.

\*The full paper for this work will be published in [1]

The process of the feature selection using F-test is consisted with following 3 steps.

- Step 1: Apply F-test to each space-time-frequency segment to calculate the F-values.
- Step 2: For each feature on space-time-frequency segment, select discriminative features which have larger F-value than the threshold value.
- Step 3: Assign a weight to each time-frequency segment with the sum of F-values of selected source signals.

### C. Classification

We train classifiers for each time-frequency segment with features of selected source signals based on Linear Discriminant Analysis (LDA) and Support Vector Machine (SVM) algorithms. In the precedent case, new input signals are transformed with the trained LDA, and classified with Gaussian distribution. The linearly weighted summation of weights and results of LDA/SVM classifiers on the time and frequency domains is used to classify motor imagery tasks.

## III. RESULTS

### A. Experimental Paradigm

We use BCI Competition IV data set II-a to demonstrate the effectiveness of the proposed classification method. EEG signals were recorded from 9 subjects about 4-class (left hand, right hand, foot, tongue) motor imagery tasks. 72 trials for each motor imagery task were recorded for training and another 288 trials were recorded for testing from each subject. We exploit all the 22 channels available. The channel location is shown in Fig. 3. Each signal was band pass-filtered between 0.5 Hz and 100 Hz and applied 50 Hz notch filter to reduce noises.

For comparison of performances in [14], We used EEG signals of the time interval, from 2.5 to 4.5 seconds, for training data. For test data, we used EEG signals in windows of the size of 2 seconds moving from 0 to 6 seconds with 10-sample intervals. Fig. 1 shows the time scheme of the experimental paradigm and extracted data in the training and test steps.

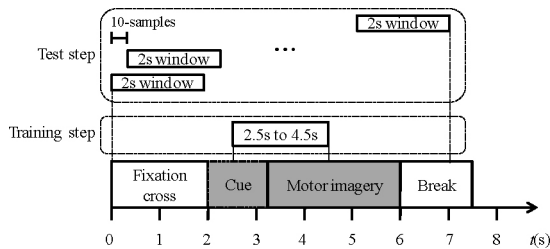


Figure 1. Time scheme of the experimental paradigm.

### B. Results of Classification

Table I shows the performance of the proposed method with LDA and SVM classification models. We use a metric of Cohen's Kappa value that measures the agreement between two estimators. The LDA model shows higher performance than the SVM model.

The classification accuracies obtained from every window used for the three competing methods are illustrated in Fig. 2. The figure clearly shows that the maximum classification accuracies are obtained by the windows positioned between 4s and 5s and there is a gradual decrease thereafter.

Table II shows the classification accuracies for 4-class motor imagery tasks. For the comparison of results with BCI competition, the accuracies are calculated by Kappa value. In these results, 6 subjects (subject 1, 3, 4, 6, 7, and 9) show higher classification accuracies in the proposed method, and the mean of accuracies for 9 subjects is highest in the proposed method with 0.60.

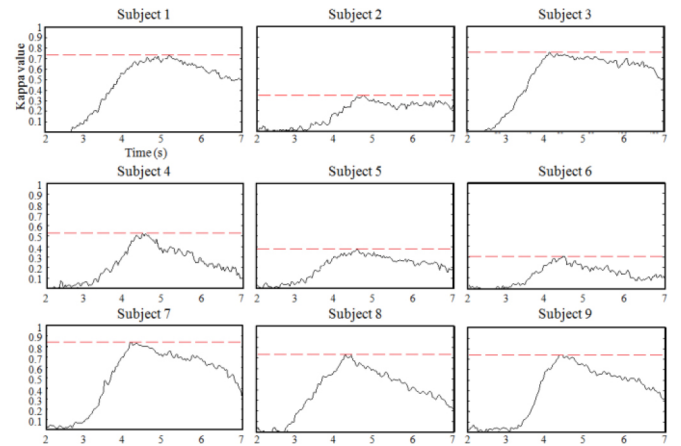


Figure 2. Time course of the Kappa values for classification performance of the test data based on the proposed method. The dotted line denotes the maximum performance of the proposed method.

TABLE I. PERFORMANCE COMPARISON OF THE PROPOSED METHOD WITH LDA AND SVM MODELS ON THE BCI COMPETITION IV DATASET II-A.

|           | Proposed method |             |
|-----------|-----------------|-------------|
|           | with LDA        | with SVM    |
| Subject 1 | 0.74            | 0.74        |
| Subject 2 | <b>0.35</b>     | 0.34        |
| Subject 3 | 0.76            | 0.76        |
| Subject 4 | 0.53            | 0.53        |
| Subject 5 | <b>0.38</b>     | 0.34        |
| Subject 6 | <b>0.31</b>     | <b>0.31</b> |
| Subject 7 | 0.84            | <b>0.86</b> |
| Subject 8 | <b>0.74</b>     | 0.72        |
| Subject 9 | 0.74            | <b>0.75</b> |
| Mean      | <b>0.60</b>     | 0.59        |

TABLE II. PERFORMANCE COMPARISON OF THE PROPOSED METHOD WITH THE WINNERS OF THE BCI COMPETITION IV DATASET II-A.

|           | Proposed Method | BCI Competition IV |            |            |
|-----------|-----------------|--------------------|------------|------------|
|           |                 | 1st winner         | 2nd winner | 3rd winner |
| Subject 1 | <b>0.74</b>     | 0.68               | 0.69       | 0.38       |
| Subject 2 | 0.35            | <b>0.42</b>        | 0.34       | 0.18       |
| Subject 3 | <b>0.76</b>     | 0.75               | 0.71       | 0.48       |
| Subject 4 | <b>0.53</b>     | 0.48               | 0.44       | 0.33       |
| Subject 5 | 0.38            | <b>0.40</b>        | 0.16       | 0.07       |
| Subject 6 | <b>0.31</b>     | 0.27               | 0.21       | 0.14       |
| Subject 7 | <b>0.84</b>     | 0.77               | 0.66       | 0.29       |
| Subject 8 | 0.74            | <b>0.75</b>        | 0.73       | 0.49       |
| Subject 9 | <b>0.74</b>     | 0.61               | 0.69       | 0.44       |
| Mean      | <b>0.60</b>     | 0.57               | 0.52       | 0.31       |

### C. Analysis of Optimized Spatial Filters and Assigned Weights

Fig. 3 shows the normalized weights of the time-frequency segments for left- and right-hand motor imageries for each subject. We can see that discriminative ERD/ERS patterns on time-frequency segments have high variances among the subjects.

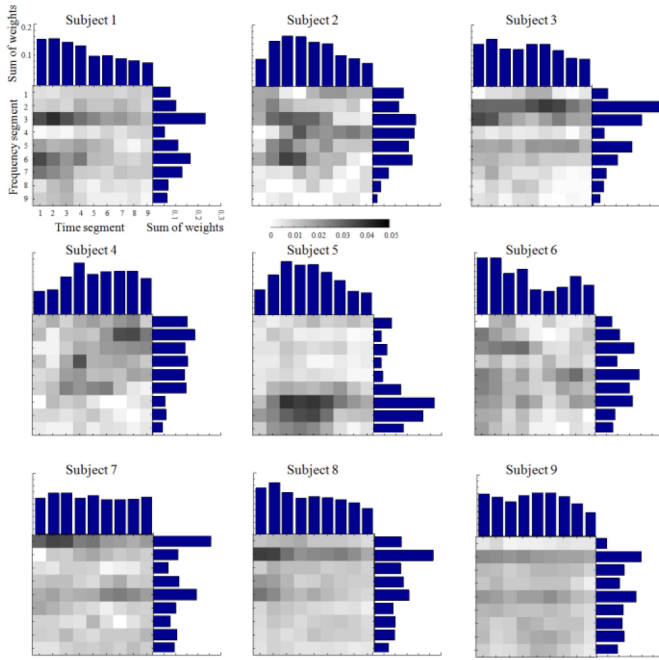


Figure 3. Normalized weights of time-frequency segments and distribution of that on timefrequency domain for left- and right-hand motor imageries.

## IV. CONCLUSION

In this paper, we proposed non-homogeneous spatial filters on the time-frequency domain for EEG based multi-class motor imagery classification. We certified that considering different sets of basis signals at different time-frequency segments leads increase of classification accuracies. While the proposed method enhanced the performance, it used a filter

bank with a fixed bandwidth dissecting a wide band of interest into smaller ones across subjects without considering spectral variation in task relevant brain responses. Therefore, it will be our forthcoming research issue.

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